

Tlotlo Oepeng

Data Scientist | Computer Vision | Geospatial ML

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Professional Summary

Data Scientist and ML Engineer with 3+ years of AI/ML experience across research, teaching, computer vision, geospatial ML, scientific ML, and Python systems used by other people. I have a physics background, a Google DeepMind-funded MSc in AI, and have worked with weather and remote-sensing datasets, adapted Prithvi-style geospatial workflows, and built ML pipelines for noisy industrial imagery at De Beers. I care most about the practical middle ground: cleaning messy data, building simple baselines, checking models carefully, and turning experiments into code that other people can use.

Selected Focus

Remote sensing	Worked with weather, climate, HLS imagery, and Prithvi-style workflows; comfortable with temporal, spatial, and physical structure in environmental datasets.
Computer vision	Builds training and evaluation workflows for noisy industrial images where blur, scale changes, and object positioning affect model behaviour.
Model testing	Uses simple baselines, subject-level splits, cross-validation, calibration checks, and failure-case analysis before trusting model results.
Engineering	Moves experiments out of notebooks with Git, FastAPI, Docker, ONNX, experiment tracking, documentation, and code review.

Technical Skills

Languages	Python, SQL, C/C++, Bash, JavaScript
ML/Data	PyTorch, TensorFlow, JAX, scikit-learn, NumPy, pandas, SciPy, OpenCV
Geospatial/Remote Sensing	Weather and climate data, HLS imagery, Prithvi workflows, geospatial time series, satellite imagery, burn-scar segmentation
Engineering	FastAPI, Docker, REST APIs, ONNX, Git, W&B, TensorBoard, Linux, repeatable pipelines
Scientific ML	Monte Carlo simulation, PINNs, transformers, anomaly detection, model evaluation

Professional Experience

ML Engineer

De Beers Group

Jan 2026 – Present

Hybrid / UK remote

- Build computer-vision training and evaluation workflows for the upcoming SynthDetect 2 machine, focused on detecting and analysing diamonds in difficult industrial imagery.
- Work with noisy, inconsistent images where blur, scale changes, and positioning make model evaluation non-trivial.
- Export PyTorch models to ONNX and test lightweight inference paths for downstream use.
- Build REST APIs around model outputs so ML results can be used by engineering workflows rather than staying in notebooks.
- Track experiments with W&B and TensorBoard, document assumptions in Confluence, and keep the codebase maintainable through Git and review.

AI Researcher / Teaching Assistant

African Institute for Mathematical Sciences, with Neil Hart, University of Oxford

2024 – 2025

South Africa

- Supported AI research and teaching at AIMS, including work around weather, climate, and geospatial machine-learning datasets.
- Helped Neil Hart deliver a course on AI for climate and remote sensing, supporting students with the practical ML and coding parts of the material.
- Customised a Prithvi Global workshop fork so students could run the geospatial model exercises on Google Cloud Platform.
- Adapted notebooks, configs, and training workflows for student use, making the material easier to set up, run, and debug in a cloud environment.
- Worked with Prithvi-style remote-sensing workflows, including HLS data and burn-scar segmentation examples.

Research Engineer, Physics-Informed ML

Florida State University collaboration with Prof. Harrison Prosper

Aug 2024 – Present

Remote

- Write Python code for cosmology PINN experiments, turning physical constraints into training runs, numerical checks, and result plots.
- Separated data generation, training loops, diagnostics, and reporting so experiments are easier to repeat and debug.
- Awarded a USD 1,200 research excellence prize for contributions connected to this collaboration.

Selected Projects

Prithvi Global Workshop Fork: Geospatial Models for Teaching

[GitHub](#)

Python, PyTorch, Prithvi, HLS data, Google Cloud Platform

- Customised a Prithvi Global workshop fork so students could run geospatial model exercises on GCP.
- Adapted notebooks, configs, and training code for classroom use, with attention to setup issues, repeatability, and debugging.
- Worked through HLS remote-sensing workflows, including burn-scar segmentation and Prithvi fine-tuning/inference examples.

GemAI - Diamond Price Intelligence API

[GitHub](#)

FastAPI, Docker, PyTorch, TabNet, pandas, Pydantic

- Built a diamond-price estimator from messy tabular data through feature cleaning, feature engineering, TabNet training, validation, and API packaging.
- Wrapped the trained model in a typed FastAPI and Docker service so predictions could be called outside the notebook.

OASIS-1 Dementia Benchmark - Leakage-Aware Medical ML

[GitHub](#)

Python, scikit-learn, PyTorch, subject-level splits, calibration, MRI CNN baseline

- Built a dementia-classification benchmark comparing clinical/morphometric baselines against a processed MRI CNN with subject-level splits to avoid leakage.
- Used cross-validation, calibration/error analysis, and baseline checks to show tabular models reach about 0.80–0.81 ROC-AUC while naive MRI models need stronger input design.

Education

MSc Mathematical Sciences (AI), University of Cape Town

2023 – 2024

Dissertation with Distinction; Google DeepMind Scholar

BSc Physics, University of Botswana

2015 – 2019

Dean's Prize; highest graduating student in the Faculty of Science

Awards, Certifications & Activities

- **Google DeepMind Scholar:** funded MSc studies in Artificial Intelligence at the University of Cape Town.
- **Research excellence:** Florida State University prize for physics-informed neural network collaboration.
- **Publication:** co-author, [Dissipative quantum fluctuations-induced droplets in Bose-Einstein mixtures](#), Physics Letters A, 2025.